Design of Novel Prosthetic Wrist Using Shape Memory Alloy Actuators and Rolling Contact Joint With sEMG Control

Chongyoung Chung, Kyujin Hyeon, Jihyeong Ma, and Ki-Uk Kyung*, *Member, IEEE*

***Abstract*— This paper proposes a novel prosthetic wrist design that replicates human wrist movements by utilizing surface electromyography (sEMG) signals for control. The design includes two rolling contact joints and shape memory alloy (SMA) spring actuators, which mimic the two-row structures of carpal bones and wrist muscles, respectively. It can perform functional ranges of motion, including 53° for flexion, 50° for extension, 40° for radial deviation, and 42° for ulnar deviation. Due to the restoring force of the SMA spring actuator, it can maintain the stiffness of the wrist even when external forces are applied, while also providing flexibility. Also, mimicking the wrist muscles enables direct mapping without complex motion detection algorithms, resulting in fast and precise control using sEMG signals. To address the cooling rate limitation of SMA actuators, a portable air compressor and nozzle structures are employed to enhance cooling efficiency. The air passing through the nozzle increases in velocity and decreases in temperature, improving the cooling rate.**

I. Introduction

Wrist motion is one of the most frequently used and important movements in the upper limbs during our activities of daily living (ADLs) [1]-[2]. It plays a vital role in tasks as grasping, manipulating objects, and performing various functional movements. Also, over 60% of the upper limb amputations are trans-carpal, replacing their wrist joint is crucial to them [2]. Therefore, various robotic prosthetic wrists have been investigated, aiming to assist individuals with upper limb amputations as well as in applications such as wearable robotic arms [3].

When designing prosthetic wrists, several key requirements need to be considered. Firstly, the wrist should be capable of performing a wide range of motions, including flexion/extension and radial/ulnar deviation, to replicate natural wrist movements [4]. Moreover, it is crucial to prioritize lightweight design to ensure comfort and ease of use for individuals or robotic users replacing their natural wrist joint. Additionally, the prosthetic wrist should enable natural and fast control based on the user's intentions [5]. One commonly used method for intention recognition is the measurement of surface electromyography (sEMG) signals that precede muscle contraction. However, traditional sEMG-based intention recognition for wrist motion can be time-consuming, mainly due to the complexity involved in implementing artificial intelligence (AI) techniques.

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C. Chung, K. Hyeon, M, J. Ma, and K.U. Kyung are with the Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology, Daejeon, 34141 South Korea (e-mail: cy.chung@kaist.ac.kr; axz0502@kaist.ac.kr; wlgudak123@kaist.ac.kr; kyungku@kaist.ac.kr).

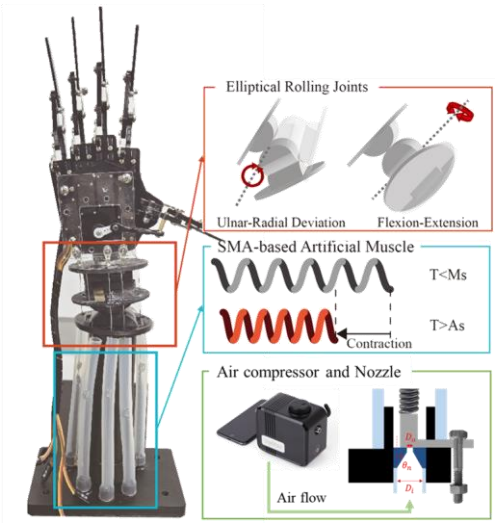


Figure 1. Overall design of the proposed prosthetic wrist

This paper introduces a novel approach to developing a lightweight and responsive prosthetic wrist, which can be controlled using surface electromyography (sEMG) signals from the muscles in the wrist. To replicate the natural range of motion (ROM) of the human wrist, we designed an elliptical rolling joint for the artificial wrist joint, and incorporated a shape memory alloy (SMA) based actuator, as depicted in Figure 1. Taking inspiration from the carpal bones' anatomy, we devised a two-row structured wrist, with each row offering a distinct degree of freedom. Additionally, by using SMA-based actuators, we can further enhance the wrist's flexibility and reduce its weight.

The movement of the wrist is achieved by mimicking the actions of six muscles: Flexor Carpi Radialis (FCR), Flexor Carpi Ulnaris (FCU), Extensor Carpi Radialis Longus (ECRL), Extensor Carpi Radialis Brevis (ECRB), Extensor Carpi Ulnaris (ECU), and Palmaris Longus (PL). Each of these muscles is emulated by an SMA actuator, chosen for its remarkable power density compared to other soft actuators [6]. However, SMA actuators do have a drawback of relatively slow actuation speed due to their cooling rate. To address this limitation, we propose the implementation of forced convection, employing a portable air compressor and a specially designed nozzle structure, to significantly improve the cooling rate.

Furthermore, surface electromyography (sEMG) signals from each wrist muscle are measured and mapped to corresponding artificial SMA muscles for activation. This approach allows a direct mapping between the artificial muscles and the muscles of the human wrist, eliminating the need for complex computational motion detection and enabling fast and precise control.

TABLE I. Selected Design Parameters of Rolling Contact Joints

Parameters	*Value (mm)*
Radius of major axis for FE ($a_{f/e}$)	14.45
Radius of minor axis for FE ($b_{f/e}$)	12.63
Height for FE ($B_{f/e}$)	11.92
Radius of major axis for UR ($a_{u/r}$)	20.04
Radius of minor axis for UR ($b_{u/r}$)	14.40
Height for UR ($B_{u/r}$)	10.30

II. Design Progress of the Artificial Wrist Joint

A. Rolling Contact Joint Design

The wrist joint's two degrees of freedom movements result from the rotational and gliding motion of each bone, combined in various ways. The two-rowed structure, as shown in Fig. 2 (a), is a well-known model explaining carpal bone movement. Based on this model, we propose a two-row structure for the rolling joint, as shown in Fig. 2 (b). The rolling joint connecting the forearm bone and proximal carpal row allows for flexion-extension motion, while the rolling joint between the proximal and distal carpal rows enables ulnar-radial deviation motion.

The rolling joints can be considered an anti-parallelogram mechanism, as shown in Fig. 2 (c). First, to design the proposed rolling contact joint, the constraint for conventional rolling joint [7], [8] expressed as

$$R \cdot sin(\frac{\theta_{req}}{n}) \leq d \tag{1}$$

where R, θ_{req}, and n are the radius of curvature, rotation angle, and number of the rolling joint, respectively, should be satisfied. Also, each rolling joint should prevent collision, another constraint [7], [8] is expressed as

$$(B-R)(1+cos(\frac{\theta_{req}}{n})) + 2Rcos(\frac{\theta_{req}}{n}) \leq \frac{L}{2}sin(\frac{\theta_{req}}{n}) \tag{2}$$

where B and L are the height and width of the rolling joint, respectively. To achieve a compact and high-load-bearing wrist joint, an optimization process was used to select suitable design parameters. The height of the rolling joint was reduced for compactness, while the curvature was decreased to lower contact stress and enhance load-bearing capacity. Since the curvature varied for the ellipse, the average value was utilized. Therefore, a cost function (C) was defined as

$$C\left(\frac{a_i}{b_i}\right) = B\left(\frac{a_i}{b_i}\right) \cdot \frac{\int_0^{\theta_{req}} \frac{1}{r_{cur}} d\theta}{\theta_{req}} (i = f/e \text{ or } u/r) \tag{3}$$

to minimize the cost function to each joint (f/e and u/r) as shown in Table I, where r_{cur} is the radius of the curvature for every point.

B. Artificial Muscle Design

We designed six artificial muscles using shape memory alloy (SMA) springs. These springs contract with increasing temperature, similar to human muscle activation. By inducing spring-like deformation in the SMA actuators, we achieved amplified output deformation, enabling movements comparable to human muscles. The coiled SMA spring actuators can generate deformation up to 300-400% [9].

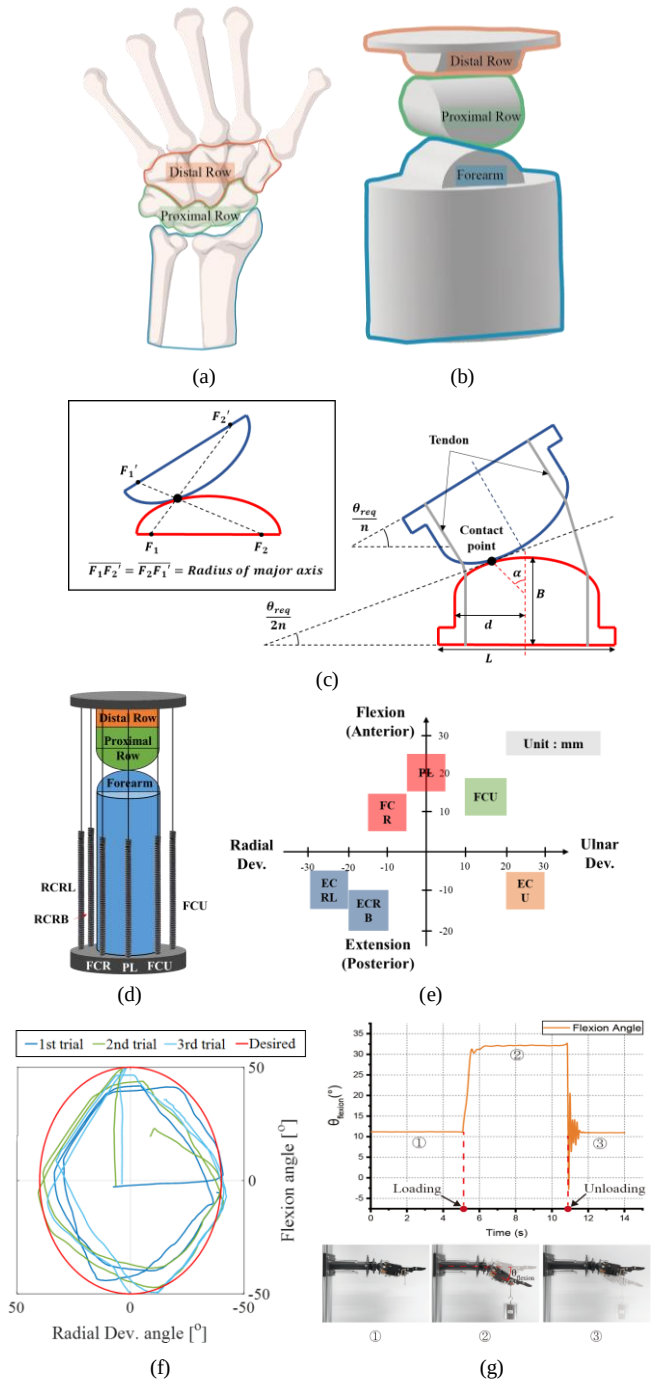


Figure 2. (a) Two-row structure of the carpal bones and (b) two-row structure of the artificial wrist joint. (c) The parameters of the elliptic rolling joint. The arrangements of the artificial muscles in (d) front and (e) top views. (f) Circumduction trajectory and (g) stability test of proposed wrist.

Various design parameters were considered for the artificial muscle, which involved selecting SMA wire with an austenite-start-temperature (A_s) slightly higher than body temperature and a martensite-start-temperature (M_s) higher than room temperature. The SMA wire diameter (d) was chosen as 0.5 mm, and the SMA spring diameter (D) was selected as 1.5 mm. Based on human muscle properties, other parameters such as the number of turns per spring (n) and the total number of springs (N), were determined.

TABLE II. Characteristics of Human Wrist Muscles

Muscles	*Moment Arm (FE)*	*Moment Arm (UR)*	*PCSA (cm^2)*	*Required Displacement*
FCR	15 mm	-10 mm	2.0	38 mm
PL	25 mm	0 mm	0.9	45 mm
FCU	20 mm	15 mm	3.2	45 mm
ECU	-10 mm	25 mm	3.8	47 mm
ECRB	-15 mm	-15 mm	2.9	38 mm
ECRL	-10 mm	-25 mm	2.1	47 mm

TABLE III. Selected Design Parameters of SMA Spring

Muscles	A_s (℃)	M_s (℃)	*n*	*N*	*d/D(mm /mm)*	*Displacement*
FCR	45.9	32.7	50	2	0.5/1.5	38 mm
PL			60	1		45 mm
FCU			60	3		45 mm
ECU			60	4		45 mm
ECRB			50	3		38 mm
ECRL			60	2		45 mm

One method to quantify muscle force is through the physiological cross-sectional area (PCSA) [10]. The PCSA represents the cross-sectional area of a muscle perpendicular to its muscle fibers, rather than its longitudinal axis. The force generated by the muscle can be described as

$$F_m = PCSA \cdot SF \cdot cos\phi \quad (4)$$

where F_m is the force generated by the muscle, SF is the specific force of the muscle, and ϕ is the pennation angle. Given that all muscles related to wrist motions are fusiform, the pennation angles are all 0. In the case of the force generated, the SMA spring can be described as

$$F_{sma} = N \cdot FS \quad (5)$$

where F_{sma} is the force generated by the muscle and FS is the force per spring. To maintain a proportional force ratio between human muscles and artificial muscles, the number of springs in the artificial muscle was determined based on the PCSA of the corresponding human muscle. The appropriate number of turns (n) for each SMA spring was calculated by considering the required displacement, following the proportionality between initial length and stretched length. Tables II and III present the characteristics of human wrist muscles and the selected design parameters. It can perform functional ranges of motion, including 53° for flexion, 50° for extension, 40° for radial deviation, and 42° for ulnar deviation as shown in Fig. 2 (f). In addition, the prosthetic wrist demonstrates high flexibility against external loads, owing to the restoring force of the SMA spring and the flexibility of the rolling joints as shown in Fig. 2 (g). This is illustrated by the ability of the proposed wrist to return to its neutral angle after a 1kg load is applied and then suddenly unloaded.

III. Air Cooling System

In the case of SMA, it can contract quickly through joule heating, but its cooling is slow, resulting in a decreased overall speed. To address this issue, a method using a coolant has been proposed, but it not only consumes a significant amount of energy but also requires a large-sized cooling circulation system [11]. Therefore, a solution was applied by using a portable air compressor (SPARMAX, Beetle compressor, weight: 0.6 kg, maximum airflow: 18 L/min, maximum pressure: 2.1 bar) to induce forced convection.

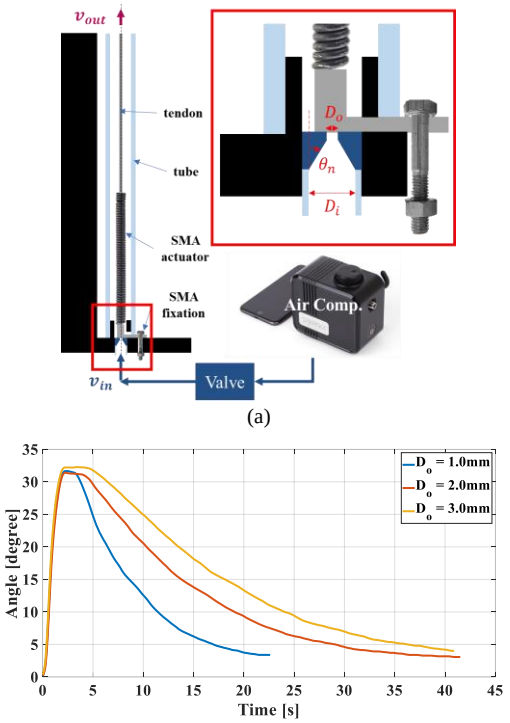


Figure 3. (a) Diagram of the cooling system composed of air compressor, nozzle, and tube. (b) Comparison of flexion angle when heating 2 seconds with 1.5A/spring and cooling.

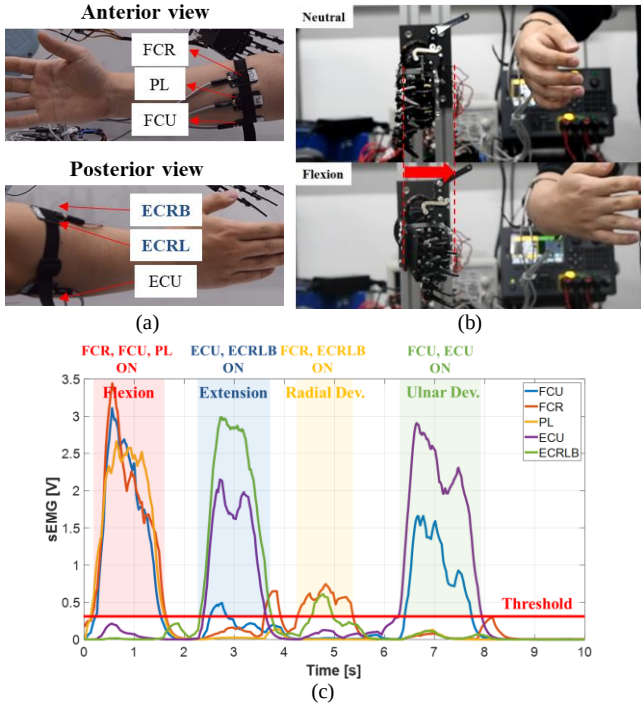


Figure 4. (a) Position of the sEMG electrodes and (b) control of proposed wrist using sEMG signals. (c) sEMG signal output during flexion/extension/radial and ulnar deviation.

To enhance the cooling effect along with the air compressor, a nozzle was used as shown in Fig. 3 (a). When passing through the nozzle, not only does the flow velocity increase, but there is also a slight decrease in temperature, which helps to increase the cooling effect [12]. The

performance of the nozzle is influenced by the inlet diameter (D_i), outer diameter (D_o), and angle of the nozzle (θ_n). However, in the system design, parameters D_i and θ_n were fixed as 4 mm and 45°, respectively. To investigate the relationship with D_o, the flexion direction actuator was heated for 2 seconds while changing D_o as 1, 2, 3 mm. Then, the valve was opened, and the corresponding actuators were cooled through each nozzle and the flexion angle was observed during this process. The results showed that the fastest cooling speed was achieved when D_o was 1mm as shown in Fig. 3 (b). Therefore, for this study, a nozzle was constructed using a 1mm as D_o value. When these designed nozzles are placed under each of the six artificial muscles, it allows for the passage of fast and cool air. Additionally, silicone tubes were used to create channels to prevent leakage and maximize the effectiveness.

IV. Control using sEMG signals

There are several methods to control the proposed prosthetic wrist, such as a joystick or command-based control. Nevertheless, this study involved the measurement of sEMG signals from the wrist, enabling the application of precise currents to individual shape memory alloy (SMA) actuators in response to these signals. As a preliminary study, the control was conducted by applying a constant current when the sEMG signal exceeded a certain threshold.

sEMG electrodes were placed on six muscle locations as shown in Fig. 4 (a) to measure signals. The measured signals underwent denoising, amplification, and filtering processes. When a signal exceeded a threshold, a current (1.5A/spring) was applied to induce contraction in the SMA muscles. During periods without current, cooling was accomplished by employing forced air convection. Control of the proposed wrist using sEMG signals is demonstrated in Fig. 4 (b). The measured signals from each muscle are shown in Fig. 4 (c), and corresponding muscles were selectively contracted based on these signals. This approach allows the control circuit to detect user intention within approximately 1 ms and apply the appropriate current, enabling quick movement of the prosthetic wrist.

V. Conclusion and Future Works

In this study, a novel prosthetic wrist was developed to replicate the human wrist anatomy. To mimick the human wrist joint and achieve a wide range of wrist motion, rolling contact joints and SMA spring actuators were utilized. In order to address the issue of slow cooling speed while maintaining a lightweight and powerful output, a proposed solution involved the combination of a nozzle with forced air convection. By incorporating a portable air compressor, the entire system weighed approximately 1kg, enabling a lightweight design. Additionally, the use of sEMG signals to measure wrist muscle activity allowed for precise control without complex AI algorithms, facilitating rapid and direct muscle matching for motion control.

However, the current nozzle design used in the study was not optimized and lacks efficiency. It is possible to achieve an optimized nozzle design that enhances cooling performance by increasing the flow velocity at the outlet for improved convection. Additionally, instead of relying solely on sEMG threshold control, there are plans to enhance force matching for each muscle by considering muscle activation and SMA contraction force based on temperature. This will involve attaching temperature sensors to each bundle of SMA muscles to monitor real-time temperature and developing control techniques to synchronize sEMG signals with SMA actuator temperature.

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